

## Supplementary data

|                                                                  |                                                                                       |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| output_directory                                                 |                                                                                       |
| aligned_reads                                                    |                                                                                       |
| <reference_id>--<sample_id>.bam                                  | : Unfiltered read alignments to references in BAM format                              |
| <reference_id>--<sample_id>.formatted.bam                        | : Quality filtered read alignments to references in BAM format                        |
| <reference_id>--<sample_id>.formatted.bam.csi                    | : Index for the filtered BAM files produced by samtools index                         |
| <reference_id>--<sample_id>.formatted.MarkDuplicates.metrics.txt | : Output from picard MarkDuplicates containing summary data tables                    |
| annotations                                                      |                                                                                       |
| <method>                                                         |                                                                                       |
| <sample_id>.embl                                                 | : Genome annotations and sequences in (multi) EMBL format                             |
| <sample_id>.faa                                                  | : CDS/ORF amino acid sequences in FASTA format                                        |
| <sample_id>.ffn                                                  | : Feature nucleotide sequences in FASTA format                                        |
| <sample_id>.fna                                                  | : Replicon/contig DNA sequences in FASTA format                                       |
| <sample_id>.gbff                                                 | : Annotations of genomes in (multi) GenBank format                                    |
| <sample_id>.gff3                                                 | : Annotations of genomes in GFF3 format                                               |
| <sample_id>.hypotheticals.faa                                    | : Hypothetical protein CDS amino acid sequences in FASTA format                       |
| <sample_id>.hypotheticals.tsv                                    | : Information on hypothetical protein CDSs in TSV format                              |
| <sample_id>.tsv                                                  | : Information for protein CDSs in TSV format                                          |
| <sample_id>.txt                                                  | : Genome annotation summary statistics in plain text format                           |
| assemblies                                                       |                                                                                       |
| flye                                                             |                                                                                       |
| <sample_id>.assembly_graph.gfa.gz                                | : Genome assembly graphs in gzipped GFA format                                        |
| <sample_id>.assembly_graph.gv.gz                                 | : Genome assembly in gzipped Graphviz format                                          |
| <sample_id>.assembly_info.txt                                    | : Statistics on contigs produced by Flye in TSV format                                |
| <sample_id>.assembly.fasta.gz                                    | : Genome assemblies in gzipped FASTA format                                           |
| <sample_id>.flye.log                                             | : Log file produced by Flye in plain text format                                      |
| <sample_id>.params.json                                          | : Parameters used to run Flye in JSON format                                          |
| spades                                                           |                                                                                       |
| <sample_id>.assembly.gfa.gz                                      | : Genome assembly graphs in gzipped GFA format                                        |
| <sample_id>.scaffolds.fa.gz                                      | : Genome assemblies in gzipped FASTA format                                           |
| <sample_id>.spades.log                                           | : Log file produced by Spades in plain text format                                    |
| <sample_id>.warnings.log                                         | : Warnings produced by Spades in plain text format                                    |
| busco                                                            |                                                                                       |
| <sample_id>--<database>-busco                                    | : BUSCO output directory                                                              |
| <sample_id>--<database>-busco.batch_summary.txt                  | : BUSCO analysis summary statistics in TSV format                                     |
| <sample_id>--<database>-busco.log                                | : Log file produced by BUSCO in plain text format                                     |
| downloads                                                        |                                                                                       |
| annotations                                                      |                                                                                       |
| <reference_id>.gff.gz                                            | : Genome annotations downloaded by the pipeline in gzipped GFF format                 |
| assemblies                                                       |                                                                                       |
| <reference_id>.fasta.gz                                          | : Genome assemblies downloaded by the pipeline in gzipped FASTA format                |
| reads                                                            |                                                                                       |
| <reference_id>_<orientation>.fastq.gz                            | : Reads downloaded by the pipeline in gzipped FASTQ format                            |
| fastp                                                            |                                                                                       |
| <sample_id>.fastp.html                                           | : FASTP report with read quality and adapter trimming statistics in HTML format       |
| <sample_id>.fastp.json                                           | : FASTP report with read quality and adapter trimming statistic in JSON format        |
| <sample_id>.fastp.log                                            | : Log file produced by FASTP                                                          |
| metadata                                                         |                                                                                       |
| reference_metadata.tsv                                           | : Reference metadata in TSV format                                                    |
| sample_metadata.tsv                                              | : Sample metadata in TSV format                                                       |
| pipeline_info                                                    |                                                                                       |
| execution_report_<date>.html                                     | : Statistics like time and ram usage for pipeline execution as a HTML report          |
| execution_trace_<date>.txt                                       | : Statistics like time and ram usage for pipeline execution in TSV format             |
| messages.tsv                                                     | : Errors, warnings, and messages produced by the pipeline in TSV format               |
| params_<date>.json                                               | : Parameters used when running the pipeline in JSON format                            |
| pathogensurveillance_run_info.yml                                | : Pipeline run metadata in YML format                                                 |
| pipeline_dag_<date>.html                                         | : Flowchart of the pathogensurveillance pipeline automatically generated by Nextflow  |
| version_info.yml                                                 | : Versions of software used in the pipeline in YML format                             |
| pirate                                                           |                                                                                       |
| <report_id>_results                                              | : Pirate output directory with gene orthology information                             |
| pocp                                                             |                                                                                       |
| <report_id>.pocp.tsv                                             | : Percentage of conserved proteins in TSV format                                      |
| quality_control                                                  |                                                                                       |
| fastqc                                                           |                                                                                       |
| <reads_id>.fastqc.html                                           | : FastQC report on read quality in HTML format                                        |
| <reads_id>.fastqc.zip                                            | : Zipped directory of FastQC output                                                   |
| multiqc                                                          |                                                                                       |
| <report_id>.multiqc                                              | : Directory of MultiQC output with intermediate files used to make the MultiQC report |
| multiqc_report.html                                              | : MultiQC report combining the summary quality control output in HTML format          |
| nanoplot                                                         |                                                                                       |
| <sample_id>.nanoplot-report.html                                 | : Nanoplot report on read quality in HTML format                                      |
| <sample_id>.NanoStats.txt                                        | : Nanoplot report on read quality in plain text format                                |
| quast                                                            |                                                                                       |
| <sample_id>.tsv                                                  | : Quast summary statistics on genome assembly quality in headerless TSV format        |
| <sample_id>                                                      |                                                                                       |
| report.html                                                      | : Quast report with data on genome assembly quality in HTML format                    |
| reference_data                                                   |                                                                                       |
| considered                                                       |                                                                                       |
| <taxon_id>.tsv                                                   | : Metadata of available assemblies on NCBI in TSV format                              |
| downloaded                                                       |                                                                                       |
| <sample_id>.tsv                                                  | : Metadata of references selected for download from NCBI in TSV format                |
| selected                                                         |                                                                                       |
| <report_id>.busco_references.tsv                                 | : IDs of references selected for multigene phylogenies in TSV format                  |
| <report_id>.core_references.tsv                                  | : IDs of references selected for multigene phylogenies in TSV format                  |
| <report_id>.mapping_references.tsv                               | : IDs of references selected for multigene phylogenies in TSV format                  |
| reports                                                          |                                                                                       |
| <report_id>.pathsurveil_report.html                              | : Primary report for pathogensurveillance in HTML format                              |
| sendsketch                                                       |                                                                                       |
| <sample_id>.txt                                                  | : Table returned by Bbmap sendsketch with initial identifications in TSV format       |
| shared_genes                                                     |                                                                                       |
| <method>                                                         |                                                                                       |
| <gene_id>.aligned.fas                                            | : Aligned sequences for each gene shared among a group of organisms in FASTA format   |
| sketch_comparisons                                               |                                                                                       |
| ani_matrices                                                     |                                                                                       |
| <report_id>.comp.csv                                             | : Estimated ANI similarity matrix made by sourmash compare in CSV format              |
| sketches                                                         |                                                                                       |
| <id>.sig                                                         | : FracMinHash signature of the given sequence made by sourmash sketch in JSON         |
| trees                                                            |                                                                                       |
| busco                                                            |                                                                                       |
| <report_id>--<cluster_id>.treefile                               | : Trees inferred from multigene phylogeny in Newick format                            |
| core                                                             |                                                                                       |
| <report_id>--<cluster_id>.treefile                               | : Trees inferred from multigene phylogeny in Newick format                            |
| snp                                                              |                                                                                       |
| <report_id>--<reference_id>.treefile                             | : Trees inferred from variant analysis in Newick format                               |
| variants                                                         |                                                                                       |
| <report_id>--<reference_id>.fasta                                | : Filtered variants in FASTA format                                                   |
| <report_id>--<reference_id>.vcf.gz                               | : Unfiltered variants in VCF format                                                   |
| <report_id>--<reference_id>.vcffilter.vcf.gz                     | : Filtered variants in VCF format                                                     |

**Supplementary Figure 1. *PathogenSurveillance* organizes and names output files for further analyses.** The left side displays the hierarchical structure of output files. Angle brackets (<>) show run specific identifiers for references, samples, and reports. The right side provides brief descriptions of each file type, highlighting their applications in downstream analyses.

|                                |                                             |                                                             | Selection criteria |                  |            |                |                     |                     |              |                |                  |            |            |              |
|--------------------------------|---------------------------------------------|-------------------------------------------------------------|--------------------|------------------|------------|----------------|---------------------|---------------------|--------------|----------------|------------------|------------|------------|--------------|
|                                |                                             |                                                             | Downloaded         | Rank Represented | Is Typical | Is Type Strain | Is RefSeq Reference | Has Species Epithet | Is Annotated | Assembly Level | Completeness (%) | Purity (%) | Contig L50 | Coverage (x) |
| Potential reference assemblies | Zophobihabitans entericus (GCF_011745665.1) | ✓                                                           | family             | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 99             | ?                | 1          | 204        |              |
|                                | Utexia brackfieldae (GCF_036251705.1)       | ✓                                                           | family             | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 99             | ?                | 1          | 624        |              |
|                                | Orbus sturtevantii (GCF_036251875.1)        | ✓                                                           | family             | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 99             | ?                | 1          | 622        |              |
|                                | Orbus wheelerorum (GCF_036251935.1)         | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 99             | ?                | 1          | 505        |              |
|                                | Gilliamella apicola (GCF_000599985.1)       | ✓                                                           | species            | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 99             | 100              | 1          | ?          |              |
|                                | Orbus mooreae (GCF_036251205.1)             | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 99             | ?                | 1          | 2007       |              |
|                                | Frischella perrara (GCF_000807275.1)        | ✓                                                           | family             | ✓                | ✓          | ✓              | ✓                   | ✓                   | G            | 98             | 100              | 1          | 70         |              |
|                                | Gilliamella intestini (GCF_900094935.1)     | ✓                                                           | genus              | ✓                | ✓          | ✓              | ✓                   | ✓                   | S            | 99             | ?                | 18         | 519        |              |
|                                | Gilliamella apicola (GCF_003201455.1)       | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | S            | 99             | 100              | 11         | 217        |              |
|                                | Gilliamella bombicola (GCF_900094945.1)     | ✓                                                           | genus              | ✓                | ✓          | ✓              | ✓                   | ✓                   | S            | 98             | ?                | 6          | 599        |              |
|                                | Orbus hercynius (GCF_003634275.1)           | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | S            | 98             | ?                | 3          | 565        |              |
|                                | Frischella perrara (GCF_003182045.1)        | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | S            | 98             | 99               | 7          | 457        |              |
|                                | Gilliamella apis (GCF_002142155.1)          | ✓                                                           | genus              | ✓                | ✓          | ✓              | ✓                   | ✓                   | Co           | 99             | 99               | 9          | 32         |              |
|                                | Orbus sasakiae (GCF_039542345.1)            | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | Co           | 99             | ?                | 1          | 49         |              |
|                                | Frischella japonica (GCF_014489845.1)       | ✗                                                           |                    | ✓                | ✓          | ✓              | ✓                   | ✓                   | Co           | 98             | ?                | 5          | 670        |              |
|                                | Gilliamella apis (GCF_030758635.1)          | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | G            | 99             | 99               | 1          | 1171       |              |
|                                | Gilliamella apicola (GCF_030758595.1)       | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | G            | 98             | 100              | 1          | 435        |              |
|                                | Gilliamella apis (GCF_030758615.1)          | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | G            | 98             | 99               | 1          | 409        |              |
|                                | Gilliamella apis (GCF_030758735.1)          | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | G            | 98             | 99               | 1          | 230        |              |
|                                | Orbus wheelerorum (GCF_036326945.1)         | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | Ch           | 99             | ?                | 1          | 607        |              |
|                                | Gilliamella apicola (GCF_046633565.1)       | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | S            | 99             | 100              | 8          | 2          |              |
|                                | Gilliamella apicola (GCF_046633605.1)       | ✗                                                           |                    | ✓                | ✗          | ✓              | ✓                   | ✓                   | S            | 99             | 100              | 12         | 3          |              |
|                                |                                             | ⋮                                                           | ⋮                  | ⋮                | ⋮          | ⋮              | ⋮                   | ⋮                   | ⋮            | ⋮              | ⋮                | ⋮          | ⋮          |              |
|                                |                                             | Gilliamella bombi (GCF_900103255.1)                         | ✓                  | genus            | ✓          | ✗              | ✓                   | ✓                   | ✓            | S              | 99               | ?          | 15         | 307          |
|                                |                                             | Gilliamella mensalis (GCF_900103085.1)                      | ✓                  | genus            | ✓          | ✗              | ✓                   | ✓                   | ✓            | S              | 98               | ?          | 6          | 400          |
|                                |                                             | Candidatus Schmidhempelia bombi str. Bimp (GCF_000471645.3) | ✓                  | family           | ✓          | ✗              | ✓                   | ✓                   | ✓            | Co             | 98               | 100        | 20         | 37           |
|                                |                                             | Gilliamella apicola SCGC AB-598-B02 (GCF_000725165.1)       | ✓                  | species          | ✓          | ✗              | ✓                   | ✓                   | ✗            | Co             | ?                | ?          | 28         | 100          |
|                                |                                             | Gilliamella apicola SCGC AB-598-I20 (GCF_000725175.1)       | ✓                  | species          | ✓          | ✗              | ✓                   | ✓                   | ✗            | Co             | ?                | ?          | 17         | 100          |
|                                |                                             | Gilliamella sp. wkB7 (GCF_001693435.1)                      | ✓                  | genus            | ✓          | ✗              | ✓                   | ✗                   | ✓            | G              | 99               | 100        | 1          | 12           |
|                                |                                             | Orbaceae bacterium ac157xtp (GCA_045014155.1)               | ✓                  | family           | ✓          | ✗              | ✗                   | ✗                   | ✓            | G              | ?                | ?          | 1          | 521          |

**Supplementary Figure 2. *PathogenSurveillance* automates downloading of references based on taxa found in the initial identification.** Representatives of taxa predicted by sendsketch are chosen based on available metadata relevant to assembly quality and downloaded. This shows the criteria used to choose references for two samples of *Gilliamella apicola*. Representatives of the species, genus, and family are chosen, in case the species-level identification is incorrect. Potential references are represented by rows and sorted from best to worst based on the selection criteria represented by columns. Selection criteria columns are sorted from most important to least important.

|                                |                                                               | ANI values for samples 305 and 379 with a range of simulated error rates |         |          |          |         |         |            |            |            |            |            |            |            |            |           |           |
|--------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------|---------|----------|----------|---------|---------|------------|------------|------------|------------|------------|------------|------------|------------|-----------|-----------|
| Potential reference assemblies |                                                               | Contextual                                                               | Mapping | 305 (NA) | 379 (NA) | 305 (0) | 379 (0) | 305 (0.01) | 379 (0.01) | 305 (0.02) | 379 (0.02) | 305 (0.03) | 379 (0.03) | 305 (0.05) | 379 (0.05) | 305 (0.1) | 379 (0.1) |
|                                | Gilliamella apicola SCGC AB-598-B02 (GCF_000725165.1) -       | ✓                                                                        | ✓       | 0.97     | 0.98     | 0.97    | 0.98    | 0.96       | 0.97       | 0.95       | 0.96       | 0.94       | 0.95       | 0.92       | 0.93       | 0.87      | 0.88      |
|                                | Gilliamella sp. wkB7 (GCF_001693435.1) -                      | ✓                                                                        | ✓       | 0.97     | 0.97     | 0.97    | 0.97    | 0.96       | 0.96       | 0.95       | 0.95       | 0.94       | 0.94       | 0.92       | 0.92       | 0.88      | 0.88      |
|                                | Gilliamella apicola (GCF_000599985.1) -                       | ✓                                                                        | ✗       | 0.95     | 0.95     | 0.95    | 0.95    | 0.94       | 0.94       | 0.93       | 0.93       | 0.92       | 0.92       | 0.91       | 0.91       | 0.86      | 0.86      |
|                                | Gilliamella apicola SCGC AB-598-I20 (GCF_000725175.1) -       | ✓                                                                        | ✗       | 0.87     | 0.88     | 0.87    | 0.88    | 0.86       | 0.87       | 0.86       | 0.86       | 0.85       | 0.85       | 0.84       | 0.84       | 0.80      | 0.78      |
|                                | Gilliamella bombicola (GCF_900094945.1) -                     | ✗                                                                        | ✗       | 0.87     | 0.87     | 0.87    | 0.87    | 0.86       | 0.86       | 0.85       | 0.86       | 0.85       | 0.85       | 0.84       | 0.83       | 0.80      | 0.79      |
|                                | Gilliamella apis (GCF_002142155.1) -                          | ✗                                                                        | ✗       | 0.87     | 0.87     | 0.87    | 0.87    | 0.86       | 0.86       | 0.85       | 0.85       | 0.85       | 0.85       | 0.84       | 0.83       | 0.79      | 0.77      |
|                                | Gilliamella mensalis (GCF_900103085.1) -                      | ✗                                                                        | ✗       | 0.87     | 0.87     | 0.87    | 0.87    | 0.86       | 0.86       | 0.85       | 0.85       | 0.85       | 0.85       | 0.84       | 0.84       | 0.79      | 0.79      |
|                                | Gilliamella bombi (GCF_900103255.1) -                         | ✗                                                                        | ✗       | 0.86     | 0.86     | 0.86    | 0.86    | 0.85       | 0.86       | 0.84       | 0.85       | 0.84       | 0.84       | 0.83       | 0.83       | 0.79      | 0.77      |
|                                | Gilliamella intestini (GCF_900094935.1) -                     | ✗                                                                        | ✗       | 0.86     | 0.86     | 0.86    | 0.86    | 0.85       | 0.86       | 0.83       | 0.85       | 0.83       | 0.84       | 0.82       | 0.83       | 0.79      | 0.00      |
|                                | Frischella perrara (GCF_000807275.1) -                        | ✓                                                                        | ✗       | 0.86     | 0.86     | 0.86    | 0.86    | 0.84       | 0.85       | 0.84       | 0.85       | 0.82       | 0.83       | 0.82       | 0.83       | 0.81      | 0.77      |
|                                | uncultured Gilliamella sp. (GCF_945271295.1) -                | ✓                                                                        | ✗       | 0.84     | 0.85     | 0.84    | 0.85    | 0.84       | 0.84       | 0.82       | 0.83       | 0.82       | 0.83       | 0.80       | 0.82       | 0.77      | 0.00      |
|                                | Zophobihabitans entericus (GCF_011745665.1) -                 | ✗                                                                        | ✗       | 0.83     | 0.82     | 0.83    | 0.82    | 0.82       | 0.81       | 0.80       | 0.81       | 0.81       | 0.81       | 0.79       | 0.79       | 0.77      | 0.00      |
|                                | Orbaceae bacterium ac157xtp (GCA_045014155.1) -               | ✓                                                                        | ✗       | 0.82     | 0.82     | 0.82    | 0.82    | 0.82       | 0.82       | 0.81       | 0.81       | 0.79       | 0.81       | 0.79       | 0.77       | 0.79      | 0.00      |
|                                | Candidatus Schmidhempelia bombi str. Bimp (GCF_000471645.3) - | ✓                                                                        | ✗       | 0.80     | 0.83     | 0.80    | 0.83    | 0.80       | 0.82       | 0.80       | 0.82       | 0.80       | 0.80       | 0.79       | 0.79       | 0.79      | 0.00      |
|                                | Orbus sturtevantii (GCF_036251875.1) -                        | ✗                                                                        | ✗       | 0.81     | 0.81     | 0.81    | 0.81    | 0.81       | 0.81       | 0.79       | 0.81       | 0.77       | 0.81       | 0.77       | 0.77       | 0.00      | 0.77      |
|                                | Utxia brackfieldae (GCF_036251705.1) -                        | ✗                                                                        | ✗       | 0.78     | 0.77     | 0.78    | 0.77    | 0.78       | 0.77       | 0.78       | 0.77       | 0.00       | 0.00       | 0.00       | 0.00       | 0.00      | 0.00      |

**Supplementary Figure 3. *PathogenSurveillance* automates selection of contextual and mapping references from those downloaded.** Nanopore reads of two samples of *Gilliamella apicola* (305 and 379) were used as test data. Short reads were simulated from the nanopore reads with a range of error rates (indicated in the parentheses in column names starting third column). *Pathogensurveillance* was run on each pair of samples from which references were selected for multigene phylogenies (contextual) and variant calling (mapping). ANI values estimated by sourmash are shown in cells and bolded ANI values represent which mapping reference was chosen for each pair of samples.